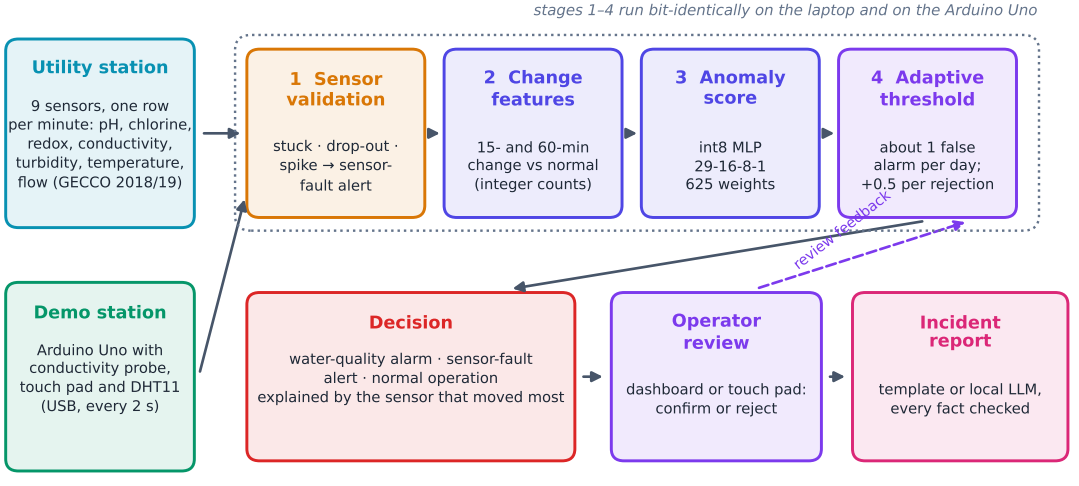
stages 1–4 run bit-identically on the laptop and on the Arduino Uno
Utility station
9 sensors, one row per minute: pH, chlorine, redox, conductivity, turbidity, temperature, flow (GECCO 2018/19)
1 Sensor validation
stuck · drop-out · spike → sensor-fault alert
2 Change features
15- and 60-min change vs normal (integer counts)
3 Anomaly score
int8 MLP 29-16-8-1 625 weights
4 Adaptive threshold
about 1 false alarm per day; +0.5 per rejection
Demo station
Arduino Uno with conductivity probe, touch pad and DHT11 (USB, every 2 s)
Decision
water-quality alarm · sensor-fault alert · normal operation
explained by the sensor that moved most
Operator review
dashboard or touch pad: confirm or reject
Incident report
template or local LLM, every fact checked
review feedback
Evaluation: time-ordered test periods · injected contaminations and sensor faults with known times · every part optional with an automatic replacement